A General Theory of Insurance

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Abstract

Classical insurance theory prices policies using expected losses, variance loadings, credibility adjustments, and capital requirements. This paper develops a **General Theory of Insurance** in which premiums are determined as functionals of individual risk, aggregate risk, actuarial uncertainty, systemic risk, model risk, and societal welfare objectives. Insurance is treated as a risk-allocation and uncertainty-minimization mechanism operating simultaneously at the levels of individuals, firms, states, and civilizations. The framework unifies actuarial science, economics, finance, statistics, machine learning, welfare economics, systems theory, and risk management into a common mathematical structure. We show that optimal premiums arise as marginal contributions to a global uncertainty functional. The resulting theory generalizes classical premium principles and provides foundations for insurance in increasingly complex and uncertain environments.

The paper ends with "The End"

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1 Introduction

Insurance is among the most important institutions ever developed by human civilization.

Its essential purpose is the transformation of uncertain losses into predictable payments.

Traditional insurance theory typically assumes:

1. Risks are measurable.
2. Probability distributions are known.
3. Losses are independently generated.
4. Sufficient diversification exists.
5. Actuarial uncertainty is negligible.

Modern reality challenges all five assumptions.

Pandemics, climate change, cyber warfare, financial crises, geopolitical conflict, demographic shifts, and technological disruptions create uncertainties extending beyond traditional actuarial frameworks.

This paper develops a general theory in which insurance becomes a mechanism for minimizing total uncertainty throughout an interconnected system.

2 The Structure of Risk

Consider a population of N insured agents.

Let

$$L = (L_1, L_2, \ldots, L_N)$$

denote the vector of losses.

Aggregate loss is

$$L_T = \sum_{i=1}^{N} L_i.$$

Classical actuarial theory focuses primarily on the distribution of L_i.

The general theory distinguishes several distinct forms of uncertainty.

2.1 Physical Risk

Physical risk arises from underlying stochastic processes.

Examples include:

- Mortality.
- Disease.
- Fire.

- Flood.
- Earthquake.
- Mechanical failure.
- War damage.

Define

$$A_i = E[\mathrm{Var}(L_i|\theta_i)]\,.$$

2.2 Actuarial Risk

Suppose losses depend upon unknown parameters:

$$L_i \sim F_i(\theta_i).$$

The insurer does not know θ_i exactly.
Using the law of total variance,

$$\mathrm{Var}(L_i) = E[\mathrm{Var}(L_i|\theta_i)] + \mathrm{Var}(E[L_i|\theta_i]).$$

Define actuarial risk:

$$U_i = \mathrm{Var}(E[L_i|\theta_i]).$$

2.3 Systemic Risk

The risk contribution of an individual depends on interactions with others.
Define

$$S_i = \mathrm{Cov}(L_i, L_T).$$

Systemic risk is especially important during:

- Financial crises.
- Pandemics.
- Wars.
- Supply-chain disruptions.
- Political instability.

2.4 Model Risk

Suppose several competing models exist:

$$F^{(1)}, F^{(2)}, \dots, F^{(K)}.$$

Define model risk by

$$M_i = \mathrm{Var}\left(E[L_i|F^{(k)}]\right).$$

Model risk reflects uncertainty regarding the correct description of reality.

3 A Unified Risk Functional

Define individual uncertainty:

$$R_i = \alpha A_i + \beta U_i + \gamma S_i + \delta M_i.$$

Aggregate uncertainty becomes

$$R_T = \rho(L_T) + \eta\Phi + \zeta\Psi,$$

where

$$\Phi = \mathrm{Var}\left(E[L_T|\Theta]\right)$$

and

$$\Psi = \mathrm{Var}\left(E[L_T|F^{(k)}]\right).$$

The insurer seeks to minimize

$$\mathcal{R} = \lambda_1 \sum_{i=1}^{N} R_i + \lambda_2 R_T.$$

4 The General Premium Functional

Traditional insurance uses

$$P_i = (1+\theta)E[L_i].$$

The general theory proposes

$$P_i = \mathcal{P}_i(A_i, U_i, S_i, M_i, R_T).$$

A first-order approximation is

$$P_i = aA_i + bU_i + cS_i + dM_i + eR_T.$$

Thus premiums depend upon both individual and collective uncertainty.

5 Marginal Risk Pricing

Define total uncertainty

$$\mathcal{R} = \lambda_1 \mathcal{R}_{\text{physical}} + \lambda_2 \mathcal{R}_{\text{systemic}} + \lambda_3 \mathcal{R}_{\text{actuarial}} + \lambda_4 \mathcal{R}_{\text{model}}.$$

Definition 1. *The marginal risk contribution of insured i is*

$$\frac{\delta\mathcal{R}}{\delta L_i}.$$

Theorem 1 (General Premium Principle)**.** *The optimal premium satisfies*

$$P_i = \frac{\delta\mathcal{R}}{\delta L_i}.$$

Proof. Suppose total premium revenue is constrained by solvency requirements.
The insurer minimizes the global uncertainty functional.
Let

$$J = \mathcal{R} + \lambda\left(\sum_i P_i - K\right).$$

First-order optimality conditions imply

$$\frac{\partial J}{\partial P_i} = 0.$$

Therefore each premium equals the marginal increase in total uncertainty generated by the corresponding insured.

Hence

$$P_i = \frac{\delta \mathcal{R}}{\delta L_i}.$$

□

6 Insurance as a Cooperative Game

Define a coalition:

$$S \subseteq N.$$

Its risk is

$$v(S) = \rho\left(\sum_{i\in S} L_i\right).$$

The Shapley allocation is

$$P_i = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(N-|S|-1)!}{N!}\left[v(S\cup\{i\}) - v(S)\right].$$

This produces a fair allocation of uncertainty costs.

7 Bayesian Insurance

Suppose prior beliefs satisfy

$$\theta_i \sim \pi(\theta_i).$$

Observed data D generate

$$\pi(\theta_i|D).$$

Expected losses become

$$E[L_i] = \int E[L_i|\theta_i]\pi(\theta_i|D)d\theta_i.$$

Actuarial uncertainty becomes

$$U_i = \int \left(E[L_i|\theta_i] - E[L_i]\right)^2 \pi(\theta_i|D)d\theta_i.$$

Improved information reduces premiums by reducing uncertainty.

8 Machine Learning and Insurance

Modern insurance increasingly relies upon:

- Deep learning.
- Gradient boosting.
- Survival models.
- Neural networks.

- Reinforcement learning.
- Generative AI.

A learning system estimates

$$\hat{L}_i = f(X_i).$$

Prediction uncertainty contributes directly to actuarial risk.

Pseudo-Code

```
Input: Data X

Estimate loss model f(X)

Estimate:
Physical Risk
Actuarial Risk
Systemic Risk
Model Risk

Compute Total Risk Functional

For each policyholder:
    Premium = Marginal Contribution
End
```

9 Biological and Medical Insurance

Biological systems generate uncertainty through:

- Genetics.
- Aging.
- Mutation.
- Epidemiology.
- Environmental exposure.

Let

$$H_i(t)$$

denote a health state.

Health insurance becomes pricing over stochastic biological trajectories.

Psychiatric and psychological risks may also enter through behavioral factors affecting future loss distributions.

10 Physics and Engineering Insurance

Engineering systems exhibit reliability structures.

Failure probabilities satisfy

$$P(\text{failure}) = 1 - P(\text{survival}).$$

For independent components,

$$P(\text{survival}) = \prod_i P_i.$$

Insurance pricing may therefore incorporate network reliability theory.

11 Political Economy and State Insurance

States themselves act as insurers.

Examples include:

- Deposit insurance.
- Unemployment insurance.
- Pension systems.
- Disaster relief.
- Sovereign guarantees.

A state effectively pools uncertainty across generations.

12 Geopolitical Insurance

Geopolitical uncertainty includes:

- Military conflict.
- Trade disruptions.
- Sanctions.
- Political instability.
- Energy insecurity.

Let

$$G_t$$

represent geopolitical stress.

Loss distributions become

$$L_i = L_i(G_t).$$

Geopolitical risk thereby enters the premium functional.

13 Warfare Economics

War generates correlated losses.

During conflict,

$$\mathrm{Cov}(L_i, L_j)$$

typically rises dramatically.

Diversification benefits collapse.

Consequently,

$$S_i = \mathrm{Cov}(L_i, L_T)$$

becomes a dominant component of premium determination.

14 Welfare Economics

Social welfare may be represented by

$$W = \sum_{i=1}^{N} u_i(c_i).$$

Insurance increases welfare by reducing consumption volatility.
The planner's problem is

$$\max W$$

subject to solvency constraints.

15 The General Optimization Problem

The insurer solves

$$\min_{P,I} \left[\lambda_1 \sum_i R_i + \lambda_2 R_T \right]$$

subject to

$$\sum_i P_i \geq E\left[\sum_i I_i\right] + K.$$

This simultaneously minimizes:

1. Individual uncertainty.
2. Aggregate uncertainty.
3. Actuarial uncertainty.
4. Model uncertainty.
5. Systemic uncertainty.

16 Conceptual Architecture

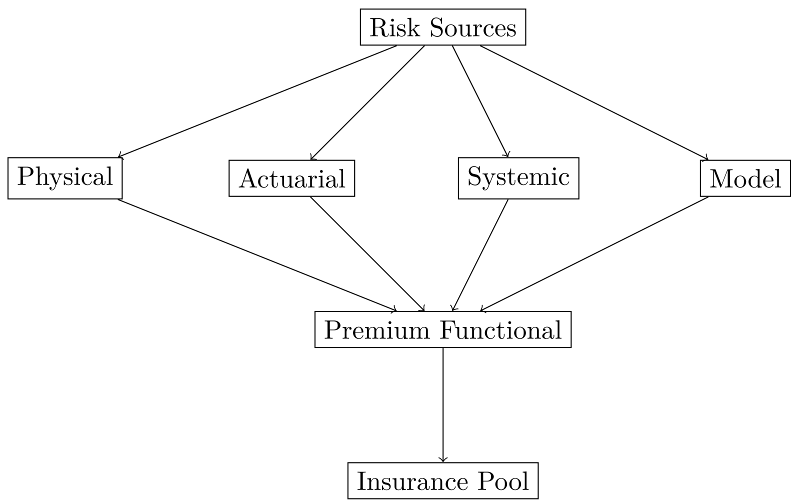

Figure 1: Conceptual architecture.

17 Comparative Summary

Theory	Risk Considered	Premium Basis
Expected Value	Mean Loss	$E[L]$
Variance Principle	Mean + Variance	$E[L] + \theta Var(L)$
Credibility Theory	Data Quality	Credibility Weights
Bayesian Theory	Parameter Risk	Posterior Losses
General Theory	Total Uncertainty	Functional Derivative

Table 1: Comparative summary.

18 Conclusion

The General Theory of Insurance views insurance as a mechanism for minimizing uncertainty throughout an interconnected system.

Premiums become functionals of:

- Physical risk.
- Actuarial risk.
- Systemic risk.
- Model risk.
- Aggregate uncertainty.

The resulting framework unifies classical actuarial science, economics, finance, machine learning, welfare economics, systems theory, and risk management within a single mathematical structure.

Insurance thus becomes not merely a transfer of losses but a coordinated reduction of uncertainty across individuals, institutions, states, and societies.

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Glossary

A_i Physical risk.

U_i Actuarial risk.

S_i Systemic risk contribution.

M_i Model risk.

R_i Individual uncertainty.

R_T Aggregate uncertainty.

P_i Insurance premium.

L_i Individual loss.

L_T Aggregate loss.

$\mathcal{R}$ Global uncertainty functional.

The End